| | |
|---|---|
| | |
| **2** | |
| **(a)** | An *ideal gas* obeys the equation $pV = nRT$ where $p$ is the gas pressure, $V$ is the volume of the gas, $n$ is the number of moles of gas, $R$ is the universal gas constant, and $T$ is the thermodynamic temperature of the gas. |
| | There are no intermolecular forces between the gas molecules, therefore an ideal gas has no potential energy. |
| **(b)(i)** | $PV = Nkt$<br>$N = \dfrac{PV}{kT}$<br>$= \dfrac{4.8\times10^{5}\times2.3\times10^{4}}{100^{3}(65+273.15)\times1.38\times10^{-23}}$ |
| | $= 2.37\times10^{24}$ (3 s.f.) |
| **(b)(ii)** | $E = \dfrac{3}{2}NkT$<br>$= \dfrac{3}{2}(1.38\times10^{-23})(65+273.15)(2.37\times10^{24})$ |
| | $= 1.66\times10^{4}$ J (3 s.f.) |

**(b)(iii)**

$$\frac{1}{2}mc_{rms}{}^{2} = \frac{3}{2}kT$$

Since $m$ and $k$ are constant,

$$c_{rms}{}^{2} \propto T$$

$$\frac{c_{rms\,65}}{c_{rms\,75}} = \sqrt{\frac{(65+273.15)}{(75+273.15)}}$$

$$= 0.986 \text{ (3 s.f.)}$$

**(c)(i)**

The product of pressure and volume of the gas at B and C are the same.

**(c)(ii)**

| process | work done on the gas / $\times 10^4$ J | heat supplied to the gas / $\times 10^4$ J | increase in internal energy / $\times 10^4$ J |
|---|---|---|---|
| $A \rightarrow B$ | −1.92 | 4.80 | 2.88 |
| $B \rightarrow C$ | 3.05 | −3.05 | 0 |
| $C \rightarrow A$ | 0 | −2.88 | −2.88 |

$$WD_{AB} = -p(\Delta V)$$

$$= -\frac{4.8\times10^{5}(6.3-2.3)\times10^{4}}{100^{3}}$$

$$= -1.92\times10^{4} \text{ J}$$

**3**